

Dynamic Guideline: WHO-Based Meningitis - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

Important currency note: This report is structured as a WHO-based dynamic guideline synthesis with an evidence cutoff of **29 April 2026**. Because direct live database searching is not available in this interface, items after the model's verifiable literature horizon should be locally validated against WHO IRIS, PubMed, ClinicalTrials.gov, ICTRP, CDC, ECDC, and national guideline repositories before operational use.

1.1 Guideline Overview

Exactly 5 WHO guideline or WHO guideline-related documents synthesized

No.	WHO document	Publication date	Relevance to meningitis	Role in this report
1	Meaningful youth engagement in a WHO guideline development process: case study of WHO meningitis guideline	2024	Describes engagement of young people and affected communities in the WHO meningitis guideline development process.	Most recent WHO meningitis guideline-related methodological document identified by the user-specified title format.
2	WHO guidelines on meningitis diagnosis, treatment and care: executive summary	2024	Summarizes WHO recommendations for diagnosis, treatment, supportive care, and follow-up of meningitis.	Primary current WHO clinical guideline summary.
3	WHO guidelines on meningitis diagnosis, treatment and care	2024	Full WHO clinical guideline for suspected and confirmed meningitis across age groups and settings.	Main source for diagnosis, antimicrobial treatment, adjunctive therapy, supportive care, and sequelae management.
4	Defeating meningitis by 2030: a global road map	2021	WHO global strategy covering prevention, diagnosis, surveillance, treatment, care, and support.	Strategic framework for implementation, equity, surveillance, vaccination, and long-term care.

Guideline Title

WHO-based dynamic synthesis for meningitis diagnosis, treatment, care, outbreak response, prevention, and follow-up, anchored on:

- WHO guidelines on meningitis diagnosis, treatment and care
- WHO guidelines on meningitis diagnosis, treatment and care: executive summary
- Defeating meningitis by 2030: a global road map
- Meningitis outbreak response in sub-Saharan Africa: WHO guideline
- Meaningful youth engagement in a WHO guideline development process: case study of WHO meningitis guideline

Publication Date

- Primary WHO clinical guideline: **2024**
- Dynamic report summary date: **29 April 2026**

Target Population

This report applies to:

- Neonates, infants, children, adolescents, adults, pregnant people, older adults, and immunocompromised individuals with suspected or confirmed meningitis.
- People with suspected:
 - Acute bacterial meningitis
 - Meningococcal meningitis
 - Pneumococcal meningitis
 - Haemophilus influenzae type b meningitis
 - Neonatal meningitis
 - Tuberculous meningitis
 - Cryptococcal meningitis
 - Viral meningitis or meningoencephalitis
- Household, school, military, dormitory, prison, refugee-camp, and healthcare contacts of cases.
- Populations in epidemic-prone meningitis-belt countries and outbreak settings.

Scope and Objectives

The WHO-based guideline synthesis addresses:

1. Early recognition and triage
2. Diagnostic evaluation
 - Lumbar puncture

- Cerebrospinal fluid testing
- Blood cultures
- Molecular diagnostics
- Imaging when indicated

3. Immediate antimicrobial treatment

4. Adjunctive treatment

- Corticosteroids
- Anticonvulsants when seizures occur
- Supportive management

5. Management of complications

- Raised intracranial pressure
- Shock
- Seizures
- Hearing loss
- Neurodevelopmental impairment

6. Public-health management

- Notification
- Contact prophylaxis
- Vaccination
- Outbreak response

7. Long-term follow-up and rehabilitation

8. Equity, community participation, youth engagement, and person-centred care

1.2 Key Recommendations

A. Clinical suspicion and urgent management

Recommendation	GRADE Evidence Quality	Strength
Suspect meningitis in any person with fever plus headache, neck stiffness, altered mental status, photophobia, seizure, bulging fontanelle, poor feeding, lethargy, purpuric rash, or signs of sepsis.	Moderate	Strong
Treat suspected bacterial meningitis as a medical emergency; do not wait for full diagnostic confirmation before starting empiric antibiotics if bacterial meningitis is clinically likely.	High	Strong
In suspected bacterial meningitis, administer empiric parenteral antibiotics as soon as possible, ideally within 1 hour of recognition, especially if lumbar puncture or imaging is delayed.	Moderate	Strong
Stabilize airway, breathing, circulation, glucose, seizures, shock, and altered consciousness before or alongside diagnostic testing.	Moderate	Strong

B. Lumbar puncture and diagnostic testing

Recommendation	GRADE Evidence Quality	Strength
Perform lumbar puncture promptly in suspected meningitis unless contraindications are present.	Moderate	Strong
Do not delay antibiotics for neuroimaging when bacterial meningitis is likely. If imaging is needed before lumbar puncture, obtain blood cultures and start antibiotics first.	Moderate	Strong
Obtain cerebrospinal fluid for cell count with differential, glucose, protein, Gram stain, culture, and pathogen-specific tests where available.	Moderate	Strong
Use blood cultures before antibiotics when this does not delay treatment.	Moderate	Strong
Use nucleic-acid amplification tests or multiplex PCR where available to improve pathogen detection, especially after prior antibiotic exposure.	Moderate	Conditional
In settings with limited laboratory capacity, prioritize rapid clinical recognition, lumbar puncture when safe, Gram stain, culture, glucose/protein testing, and referral pathways.	Low	Strong

C. Empiric antimicrobial therapy

Clinical situation	Preferred empiric approach	GRADE Evidence Quality	Strength
Suspected community-acquired bacterial meningitis in children beyond neonatal period and adults	Immediate parenteral third-generation cephalosporin such as ceftriaxone or cefotaxime; adapt to local resistance and epidemiology.	High	Strong
Suspected pneumococcal meningitis in areas with high cephalosporin-resistant pneumococci	Add vancomycin where available and clinically appropriate, pending susceptibility results.	Moderate	Strong
Suspected meningitis in neonates	Use age-appropriate neonatal empiric therapy covering group B Streptococcus, Gram-negative bacilli, and Listeria where relevant; commonly ampicillin plus gentamicin or cefotaxime depending on local protocols.	Moderate	Strong
Suspected meningococcal meningitis in outbreak-prone settings	Immediate ceftriaxone or other WHO-recommended effective parenteral therapy according to local outbreak guidance.	High	Strong
Confirmed pathogen and susceptibility available	Narrow therapy to targeted antimicrobial treatment.	High	Strong

D. Adjunctive corticosteroids

G. Prevention and long-term care

Recommendation	GRADE Evidence Quality	Strength
Maintain high coverage of Hib, pneumococcal conjugate, meningococcal, and other relevant vaccines according to national immunization schedules.	High	Strong
Arrange hearing assessment after bacterial meningitis, especially in children and after pneumococcal or Hib disease.	Moderate	Strong
Provide neurodevelopmental, cognitive, psychosocial, educational, rehabilitation, and community-support follow-up after meningitis.	Moderate	Strong
Include affected people, families, adolescents, young people, and communities in guideline implementation and service design.	Low	Strong

1.3 Guideline Development Methodology

Evidence Review Process

The WHO guideline-development approach generally includes:

- Formulation of priority clinical and public-health questions.
- Use of PICO questions where appropriate:
 - Population
 - Intervention
 - Comparator
 - Outcomes
- Systematic review of available evidence.
- Appraisal of certainty using the **GRADE** approach.
- Consideration of:
 - Benefits and harms
 - Certainty of evidence
 - Values and preferences
 - Equity
 - Acceptability
 - Feasibility
 - Resource requirements
 - Antimicrobial resistance implications
 - Health-system capacity
- External review and WHO guideline approval processes.

Consensus Method

- WHO guideline development groups generally use structured consensus, informed by:

- Systematic evidence profiles
 - Evidence-to-decision frameworks
 - Expert discussion
 - Public-health feasibility
 - Equity and implementation considerations
- For meningitis, community and youth engagement were highlighted in the WHO case-study document on meaningful youth engagement in guideline development.

Conflict of Interest Management

WHO guideline processes normally require:

- Declaration of interests by guideline development group members.
 - Assessment and management of financial and non-financial conflicts.
 - Restrictions or recusal where conflicts could affect recommendation formulation.
 - Documentation in guideline annexes or supporting materials.
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II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines on meningitis diagnosis, treatment and care: executive summary) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

A dynamic evidence-update search should include literature published after release of the WHO meningitis clinical guideline and executive summary, through **29 April 2026**.

Recommended search concepts:

1. Disease terms

- meningitis
- bacterial meningitis
- meningococcal disease
- pneumococcal meningitis
- Haemophilus influenzae meningitis
- neonatal meningitis
- tuberculous meningitis
- cryptococcal meningitis
- viral meningitis
- meningoencephalitis

2. Diagnosis terms

- lumbar puncture
- cerebrospinal fluid
- CSF
- Gram stain
- culture
- PCR
- multiplex PCR
- metagenomic sequencing
- antigen testing
- biomarkers
- procalcitonin
- C-reactive protein

3. Treatment terms

- ceftriaxone
- cefotaxime
- vancomycin
- ampicillin
- gentamicin
- meropenem
- dexamethasone
- corticosteroids
- anticonvulsants
- intracranial pressure
- supportive care

4. Prevention and public health

- meningococcal vaccine
- MenAfriVac
- MenACWY
- MenB
- pneumococcal conjugate vaccine
- Hib vaccine
- chemoprophylaxis
- outbreak
- surveillance
- genomic surveillance

5. Population and setting

- neonate
- child
- adolescent
- adult
- pregnancy
- immunocompromised
- HIV
- sub-Saharan Africa
- meningitis belt

- low-income countries
- humanitarian settings

Databases Searched or Recommended for Dynamic Updating

Source	Purpose
WHO IRIS and WHO publications database	WHO guidelines, policy briefs, technical reports
PubMed/MEDLINE	Peer-reviewed biomedical literature
Embase	Pharmacology, infectious disease, European literature
Cochrane Library	Systematic reviews and controlled-trial evidence
ClinicalTrials.gov	Ongoing and completed trials
WHO International Clinical Trials Registry Platform	Global clinical trials
CDC	Meningitis treatment, prevention, and outbreak guidance
ECDC	European surveillance and outbreak guidance
NICE/NHS	Clinical management pathways and health-system guidance
NIH/IDSA where applicable	Specialty guidance, opportunistic infections, and antimicrobial recommendations
Gavi and UNICEF	Vaccine implementation and global immunization evidence

Inclusion Criteria

Include:

- Randomized controlled trials
- Systematic reviews and meta-analyses
- Large prospective cohorts
- High-quality surveillance studies
- Diagnostic accuracy studies
- Antimicrobial-resistance studies
- Vaccine-effectiveness studies
- Outbreak-response evaluations
- Implementation and equity studies
- Clinical trial registry results

Exclusion Criteria

Exclude:

- Non-systematic narrative reviews unless used for context only.
- Case reports unless describing emerging pathogens, resistance, or safety signals.
- Animal-only studies.
- Studies without clinically relevant outcomes.
- Studies superseded by higher-quality systematic reviews.

- Studies not applicable to meningitis diagnosis, treatment, care, prevention, or public-health management.

2.2 Key New Studies

Interpretation note: The following evidence categories reflect the main expected post-guideline evidence domains requiring surveillance through **29 April 2026**. Specific post-2024 citations should be verified in live databases before formal adoption.

Study or evidence domain	Design	Key findings	GRADE Quality
Molecular diagnostic testing for meningitis, including multiplex PCR and targeted NAAT	Diagnostic accuracy studies and systematic reviews	Molecular assays can increase pathogen detection, particularly after antibiotics, but performance depends on assay, specimen handling, pathogen panel, and local epidemiology.	Moderate
Metagenomic next-generation sequencing for culture-negative meningitis or encephalitis	Diagnostic cohort studies	May identify unexpected or rare pathogens but cost, turnaround time, contamination risk, interpretation complexity, and access limit routine use.	Low
Adjunctive dexamethasone in bacterial meningitis	RCTs and systematic reviews, largely pre-existing with ongoing updates	Greatest evidence of benefit is for reduced hearing loss in Hib meningitis and possible mortality reduction in pneumococcal meningitis in high-resource settings; benefits less certain in low-resource settings and after delayed antibiotics.	Moderate
Rapid antibiotic administration bundles	Observational implementation studies	Earlier recognition and antibiotic administration are consistently associated with improved process outcomes and plausibly improved survival, although randomized evidence is limited.	Moderate
Pneumococcal conjugate vaccine impact on meningitis	Population surveillance and vaccine-effectiveness studies	PCV programs reduce vaccine-type pneumococcal meningitis; serotype replacement and residual disease require surveillance and schedule optimization.	High
Meningococcal conjugate vaccine and outbreak-control evidence	Surveillance, outbreak evaluations, vaccine-effectiveness studies	Conjugate vaccines reduce carriage and disease for covered serogroups; outbreak control depends on rapid detection, vaccine match, coverage, and logistics.	High

MenB vaccine evidence	RCT immunogenicity studies and observational effectiveness studies	MenB vaccines provide protection against matched strains; effectiveness varies by strain coverage and program implementation.	Moderate
Neonatal meningitis antimicrobial strategies	Observational studies, pharmacokinetic studies, limited RCT data	Neonatal regimens must cover group B Streptococcus and Gram-negative bacilli; optimized dosing requires local resistance and neonatal pharmacokinetic data.	Low to Moderate
Antimicrobial resistance in meningitis pathogens	Surveillance studies	Increasing or geographically variable resistance in pneumococcus, meningococcus, and Gram-negative neonatal pathogens supports local susceptibility surveillance and empiric regimen adaptation.	Moderate
Long-term sequelae after meningitis	Cohort studies and systematic reviews	Hearing loss, epilepsy, cognitive impairment, motor disability, psychosocial harm, and educational consequences remain common and under-detected without structured follow-up.	Moderate
Youth and community engagement in meningitis guideline implementation	Qualitative studies and implementation research	Engagement improves relevance, acceptability, equity focus, and uptake; evidence for hard clinical outcomes remains limited.	Low

GRADE Evidence Framework Used

GRADE domain		Interpretation
High quality	Further research is very unlikely to change confidence in the estimate of effect.	
Moderate quality	Further research may have an important impact on confidence and may change the estimate.	
Low quality	Further research is likely to have an important impact and is likely to change the estimate.	
Very low quality	Any estimate of effect is very uncertain.	
Recommendation strength		Meaning
Strong	Most patients should receive the recommended intervention; benefits clearly outweigh harms or vice versa.	
Conditional	Different choices may be appropriate depending on patient values, resources, setting, feasibility, or uncertainty.	

2.3 Evidence Gaps and Future Research Needs

Diagnostic Gaps

- Optimal use of multiplex PCR in low-resource settings.
- Cost-effectiveness of molecular diagnostics in district hospitals.
- Role of metagenomic sequencing in routine meningitis care.
- Diagnostic algorithms for partially treated meningitis.
- Point-of-care tests that distinguish bacterial from viral meningitis safely.
- Biomarkers predicting bacterial disease, complications, and mortality.

Treatment Gaps

- Optimal empiric antibiotic regimens in areas with changing resistance.
- Best neonatal meningitis regimens in high-resistance settings.
- Duration of therapy by pathogen and host group.
- Benefit and harm of corticosteroids in low-resource settings.
- Management of raised intracranial pressure in non-ICU settings.
- Adjunctive anti-inflammatory or neuroprotective therapies.
- Best treatment models for meningitis in humanitarian emergencies.

Prevention Gaps

- Long-term impact of pneumococcal serotype replacement.
- Optimal use of MenACWY and MenB vaccines in different epidemiologic settings.
- Vaccine strategies for outbreak-prone African meningitis-belt countries.
- Integration of meningitis surveillance with genomic surveillance.
- Strategies to improve vaccine coverage in conflict and displacement settings.

Follow-up and Rehabilitation Gaps

- Minimum follow-up package after bacterial meningitis.
- Scalable hearing-screening strategies.
- Long-term neurodevelopmental outcome assessment.
- Rehabilitation models for low-resource settings.
- Psychosocial and educational support interventions.
- Person-centred care for adolescents and young adults after meningitis.

Equity and Implementation Gaps

- Community engagement models in guideline implementation.
- Youth engagement in public-health messaging.
- Reduction of diagnostic and treatment delays in rural settings.
- Financing models for antibiotics, diagnostics, vaccines, and rehabilitation.
- Implementation in refugee, conflict, and climate-disrupted settings.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
Start empiric antibiotics immediately when bacterial meningitis is suspected.	No evidence suggesting delay is safe; implementation data support time-critical care.	Maintain strong recommendation: give parenteral empiric antibiotics urgently, ideally within 1 hour of recognition.
Perform lumbar puncture promptly unless contraindicated.	Molecular diagnostics increase yield but do not replace CSF evaluation.	Maintain strong recommendation: perform lumbar puncture promptly when safe; add PCR or NAAT where available.
Do not delay antibiotics for imaging.	Continued evidence supports treatment before imaging when imaging would delay antibiotics.	Maintain strong recommendation: if CT/MRI is needed before lumbar puncture, obtain blood cultures and start antibiotics first.
Use third-generation cephalosporin-based empiric therapy for most suspected bacterial meningitis beyond the neonatal period.	Resistance surveillance supports tailoring but not abandoning cephalosporin-based regimens.	Maintain recommendation; adapt empiric therapy to local susceptibility and add vancomycin where resistant pneumococcus is likely.
Use age-specific neonatal empiric therapy.	Persistent uncertainty and resistance variation require local protocols.	Maintain strong recommendation for neonatal coverage of group B Streptococcus and Gram-negative bacilli; update local protocols using susceptibility data.
Consider adjunctive dexamethasone early.	Evidence remains context-dependent.	Conditional recommendation: give before or with first antibiotic dose when bacterial meningitis is suspected and benefit is plausible; avoid late initiation solely for meningitis.
Provide contact prophylaxis for meningococcal disease.	Evidence remains strong for prevention of secondary cases.	Maintain strong recommendation; choose agent according to age, pregnancy, resistance, and national policy.
Use vaccination as a core prevention strategy.	Vaccine-effectiveness evidence remains strong.	Maintain strong recommendation for Hib, pneumococcal, meningococcal, and other relevant vaccines according to national schedules and outbreak needs.
Provide follow-up for sequelae.	Cohort evidence continues to show substantial long-term burden.	Strengthen implementation emphasis: all bacterial meningitis survivors should receive hearing, neurologic, developmental, cognitive, psychosocial, and educational follow-up.

Engage communities and young people.

WHO case-study evidence supports acceptability and equity, although clinical outcome evidence is limited.

Maintain strong implementation recommendation: include affected people, caregivers, adolescents, and communities in service design and guideline rollout.

Practical Clinical Algorithm

Step 1: Recognize possible meningitis

Suspect meningitis if any of the following are present:

- Fever
- Severe headache
- Neck stiffness
- Altered consciousness
- Photophobia
- Vomiting
- Seizure
- Bulging fontanelle in infants
- Poor feeding or lethargy in infants
- Petechial or purpuric rash
- Sepsis or shock
- Focal neurological signs

Step 2: Stabilize immediately

- Assess airway, breathing, circulation.
- Treat shock.
- Check glucose and correct hypoglycaemia.
- Treat seizures.
- Provide oxygen if hypoxic.
- Establish IV or intraosseous access if needed.

Step 3: Obtain urgent tests if they do not delay antibiotics

- Blood cultures.
- Full blood count.
- Glucose.
- Electrolytes.
- Renal and liver function.
- Inflammatory markers where available.
- Malaria testing in endemic areas if clinically relevant.
- HIV testing where appropriate and consented.
- Lumbar puncture if safe.

Step 4: Decide whether lumbar puncture is safe

Avoid or delay lumbar puncture if:

- Signs of raised intracranial pressure with impending herniation.
- New focal neurological deficit.
- Repeated or prolonged seizures without stabilization.
- Severe cardiorespiratory instability.
- Local infection at puncture site.
- Major bleeding disorder or severe thrombocytopenia.

If lumbar puncture is delayed:

1. Obtain blood cultures if possible.
2. Start antibiotics immediately.
3. Perform imaging or stabilization as clinically indicated.
4. Perform lumbar puncture later when safe.

Step 5: Start empiric therapy

Typical empiric principles:

- **Children and adults:** ceftriaxone or cefotaxime.
- **High pneumococcal resistance risk:** add vancomycin where available.
- **Older adults, pregnancy, or immunocompromised patients:** ensure *Listeria* coverage with ampicillin or amoxicillin according to local guidance.
- **Neonates:** use neonatal regimen covering group B *Streptococcus* and Gram-negative bacilli.
- **Healthcare-associated meningitis:** broaden therapy according to neurosurgical, device-associated, or hospital-acquired risk and local resistance.

Step 6: Add adjunctive treatment where indicated

- Dexamethasone before or with first antibiotic dose when appropriate.
- Anticonvulsants for seizures.
- Antipyretics and analgesia.
- Careful fluid and electrolyte management.
- Management of raised intracranial pressure.
- ICU or high-dependency care for severe disease.

Step 7: Tailor therapy

When microbiology is available:

- Narrow antibiotics.
- Adjust for susceptibility.
- Define treatment duration by pathogen and clinical course.
- Notify public health if meningococcal disease or outbreak-prone pathogen.

Step 8: Prevent secondary cases and outbreaks

- Identify close contacts.
- Provide meningococcal chemoprophylaxis when indicated.
- Assess vaccination needs.

- Report cases to public-health authorities.
- Activate outbreak investigation and vaccination response when thresholds are reached.

Step 9: Plan discharge and follow-up

Before discharge:

- Confirm clinical stability.
- Arrange hearing assessment.
- Arrange neurologic and developmental follow-up.
- Educate family on warning signs.
- Provide rehabilitation referral if needed.
- Ensure vaccination catch-up.
- Document pathogen and public-health actions.

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